

Analysis Tutorial 4b

Problem 1b

Since f is bounded, $|f(x)| \leq M$ for some $M \in \mathbb{R}$, for all $x \in [a, b]$. So $-M \leq f(x) \leq M$, and since $-M$ and M are integrable, so is f . Partition $[a, b]$ as follows: $x_0 = a$

$$x_m = a + m \cdot \frac{b-a}{n} \text{ for fixed } n.$$

Then $a = x_0 < x_1 < x_2 < \dots < x_n = b$

Define $U(x) = \sup_{x_{k-1} \leq t \leq x_k} f(t)$ for $x \in [x_{k-1}, x_k]$.

Clearly $f(x) \leq U(x)$. Since U has finite values, it is measurable.

$$\text{Now } \int_{[a,b]} f d\lambda = \int f \chi_{[a,b]} d\lambda$$

$$= \int f \left(\sum_{k=1}^n \chi_{[x_{k-1}, x_k]} \right) d\lambda$$

$$= \sum_{k=1}^n \int_{[x_{k-1}, x_k]} f d\lambda$$

$$\leq \sum_{k=1}^n \int_{[x_{k-1}, x_k]} U d\lambda$$

constant on $[x_{k-1}, x_k]$

$$= \sum_{k=1}^n (x_k - x_{k-1}) \cdot U(x_{k-1})$$

$$= U(f, P).$$

Clearly $\int_a^b f(x) dx \geq L(f, P)$.

$$\text{So } \left| \int_{[a,b]} f d\lambda - \int_a^b f(x) dx \right| \leq U(f, P) - L(f, P)$$

$$\rightarrow 0 \text{ as } n \rightarrow \infty$$

Problem 17

$$\begin{aligned}\int_{[a,b]} f d\lambda &= \int_{\mathbb{Q}^c \cap [a,b]} f d\lambda + \int_{\mathbb{Q} \cap [a,b]} f d\lambda \\ &= \int_{\mathbb{Q}^c \cap [a,b]} 1 d\lambda + \int_{\mathbb{Q} \cap [a,b]} 0 d\lambda \\ &= \lambda(\mathbb{Q}^c \cap [a,b]) + 0 \\ &= \lambda([a,b]) - \lambda(\mathbb{Q} \cap [a,b]) \\ &= (b-a) - 0 = b-a\end{aligned}$$

So f is Lebesgue integrable.

On the other hand, let $\varepsilon = \frac{1}{2}$. For any $\delta > 0$, partition $[a,b]$ into $\lceil \frac{b-a}{\delta} \rceil = m$ intervals of at most δ (i.e., $x_0 = a, x_1 = a + \delta, \dots, x_{m-1} = a + (m-1)\delta, x_m = b$, with $x_{m-1} - x_{m-2} = b - a - (m-2)\delta = b - a - m\delta + \delta \leq \delta$).

But for any interval $[x_{k-1}, x_k]$, $\sup_{x_{k-1} \leq x \leq x_k} f(x) = 1$ and $\inf_{x_{k-1} \leq x \leq x_k} f(x) = 0$.

So $U(f, P) - L(f, P) \geq 1$ so f is not Riemann integrable.

Problem 18

First note that $\left| \frac{\sin x}{x} \right| \leq \frac{|x|}{|x|} = 1$ for $x \neq 0$.

For $x=0$, $f(0)=1$. So $|f|$ is bounded by 1, and thus on $[0, a]$, f is integrable. (f is measurable as $\lambda([0, a]) = a$ which is finite)

On the other hand, consider

$$A = \left\{ x \in [0, \infty) : \sin x \geq \frac{1}{2} \right\} = \bigcup_{n \in \mathbb{N}} \left[\frac{\pi}{3} + 2\pi n, \frac{2\pi}{3} + 2\pi n \right].$$

$$\text{Then } \int_{[0, \infty)} f^+ d\lambda \geq \int_A f d\lambda \quad := [a_n, b_n]$$

$$= \sum_{n \in \mathbb{N}} \int_{[a_n, b_n]} \frac{\sin x}{x} d\lambda$$

$$\geq \frac{1}{a_n}$$

$$\geq \sum_{n \in \mathbb{N}} \frac{1}{a_n} \lambda([a_n, b_n]) = \sum_{n \in \mathbb{N}} \frac{\pi/3}{\pi/3 + 2\pi n} = \infty$$

So f is not Lebesgue integrable on $[0, \infty)$

Since f is continuous on $[0, a]$ and $|f|$ bounded, it is Riemann integrable.

$$\text{So } \lim_{a \rightarrow \infty} \int_{[1, a]} f d\lambda = \lim_{a \rightarrow \infty} \int_1^a f(x) dx$$

$$= \lim_{a \rightarrow \infty} \int_1^a \frac{\sin x}{x} dx$$

$$= \lim_{a \rightarrow \infty} \left[-\frac{\cos x}{x} \Big|_1^a + \int_1^a \frac{\cos x}{x^2} dx \right]$$

$$\rightarrow -\cos 1$$

$$\text{if } \int_1^a \frac{1}{x^2} dx \rightarrow \frac{1}{2} \text{ as } a \rightarrow \infty$$

So this integral is finite